

D3), two 5mm LEDs (LED1 and LED2) and a few other components.

Input DC voltage of 3.7V is applied to connector CON1. The circuit works with DC voltage in the range of 3.0V to 40.0V. IC1 works as a step-up converter to produce 6V output voltage (V_{OUT1}), which is available

at connector CON2. For Schottky diode D1, you can use any of 1N5817, 1N5818 and 1N5819 variants.

Diode D3 reduces the 6V output voltage to around 5.3V (V_{OUT2}), which is available across connector CON3. Schottky diode D4 further reduces 5.3V to around 5.0V (V_{OUT3}), which is available across connector CON4.

The total maximal output current from all outputs at CON2, CON3 and CON4 is around 100mA, which can be adjusted with the help of components in the circuit. The DC-to-DC converter also outputs around 15mV ripples, which can be reduced to some degree with the help of capacitor C3 and additional LC or RC filters.

Construction and testing

An actual-size PCB layout for the 3.7V to 5V and 6V DC-to-DC converter is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable box. Fix connector CON2 through CON4 and LED1 and LED2 on front side of the box.

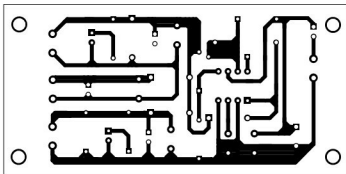


Fig. 2: Actual-size PCB layout of the DC-to-DC converter

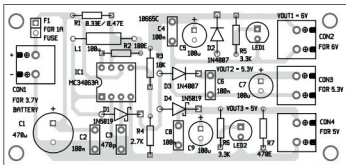


Fig. 3: Components layout for the PCB

The circuit does not need any adjustment. But you need to check input and output voltages, currents and frequency of operation of IC1. It is also good to control the current of inductor L1.

The circuit can be tested with a 3.7V Li-ion battery or with three 1.2V NiCd or NiMH batteries. You can also use any regulated power supply source producing 3.7V.

Note. 1. The output voltage without load depends on the value of resistor R7 (between 270 and 470 ohms).

2. You should always have a load connected at CON4, at least 10mA, in order to allow normal operation of the circuit.

3. Current sensing mainly depends on resistor R1. **EFY**



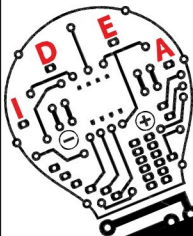
Petre Tzv Petrov was a researcher and assistant professor in Technical University of Sofia (Bulgaria) and expert-lecturer in OFPPT (Casablanca), Kingdom of Morocco. Now he is working as an electronics engineer in the private sector in Bulgaria.

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